Fiscal Year:	FY 2021	**Task Last Updated:**	FY 12/28/2020
PI Name:	T'ien, James S. Ph.D.		
Project Title:	Flame Spread and Extinction Over Solid Materials (Growth and Extinction Limit, GEL)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	44106-1712	**Congressional District:**	11
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2009 Combustion Science NNH09ZTT001N
Start Date:	03/01/2016	**End Date:**	02/28/2022
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Olson, Sandra	**Contact Phone:**	216-433-2859
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 2/28/2022 per PI (Ed., 1/22/21)		
Key Personnel Changes/Previous PI:	December 2020: Dr. Michael Johnston, USRA, is now a CoInvestigator.		
COI Name (Institution):	Ferkul, Paul Ph.D. (NASA Glenn Research Center/ National Center for Space Exploration Research) Olson, Sandra Ph.D. (NASA Glenn Research Center) Johnston, Michael C Ph.D. (Universities Space Research Association)		
Grant/Contract No.:	NNX16AE84A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Continuation of "Flame spread and extinction over solid materials," grant NNX10AD99A. Flammability of solid materials is of interest to NASA as a first line of defense in spacecraft fire safety efforts. In micro- or partial-gravity, the reduced buoyant flows, coupled with low-speed air circulation currents, create an environment quite different than that in normal Earth gravity. In addition, there are several oxidizer atmospheres under consideration for crew vehicles and habitats, which expand the scope of the problem. A fundamental understanding of solid flammability as a function of gravity, flow velocity, oxygen percentage, pressure, and sample configuration can be an important contribution both to combustion science and to the NASA space exploration initiatives. This proposed experiment will concentrate on the flame growth, decay, and extinction over the surface of a non-flat thick solid in microgravity. In particular, a solid sphere of substantial size (i.e., 4 cm diameter) is chosen as a representative of non-flat samples. In addition to the parameters influencing the flammability of thin solids, the degree of interior heat-up is an important parameter on the solid burning characteristics of thick specimens. For spherical samples, the degree of interior heating is always changing. The problem is therefore unsteady in nature. In addition, flow around a sphere is different from that around a flat surface. The existence of a forward stagnation point, shoulder and wake regions result in a different local flow pattern, hence a different flame-solid interaction. These can affect the burning and extinction characteristics. In the proposed experiment, cast Polymethylmethacrylate (PMMA) spheres will be instrumented with five embedded thermocouples to record the interior temperatures during the preheating and the combustion processes. The project objectives are (1) Experimentally determine the flame growth characteristics (growth rate, flame shape, and dimensions) over a thick solid fuel as a function of flow velocity, oxygen percentage, pressure, and the degree of internal heating; (2) Experimentally determine the flame extinction characteristics (quenching and blowoff limits) over a thick solid fuel as a function of flow velocity, oxygen percentage, pressure, and the degree of internal heating; and (3) Establish a high-fidelity numerical model that can be compared with the microgravity results and serve as a tool connecting normal gravity and microgravity performance. The proposed experiment is to be conducted aboard the International Space Station (ISS) in the Combustion Integrated Rack (CIR) because ground-based microgravity facilities cannot provide the test time duration needed for thick samples.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This is a fundamental study on the flammability of solids. The understanding to be achieved will help us to deal with fire safety on Earth and in space.
Task Progress:	1. PMMA sample fabrication. Last year, 30 thermocouple-instrumented PMMA spheres were ordered from our vendor, Meisner Galley, Inc. at New York state. Out of these, 15 were not made properly. The vendor promised to re-make 15 samples free of charge. But, we need to prepare again the pre-cast kits ourselves without ZINN's support. This year, the order was further delayed by the shutdown of vendor's lab due to COVID-19. We eventually received the re-made samples in August, 2020. Out of the 15 re-cast samples, only 9 are deemed good. Since these are used as the spared samples for 1-g testing, we felt 9 were adequate. We also ordered 30 un-instrumented PMMA sphere samples for ground testing (cheaper to make). The newly received instrumented samples have been CT-scanned to precisely determine the thermocouple positions. 2. Ignition Test. Properly controlled ignitions are critical to the success of the Growth and Extinction Level (GEL) space experiment. Many parameters need to be quantified in the proposed ignition process using heated hot wires. A systematic experimental study has been carried out to investigate the igniter current level, igniter-sample separation distance, flow velocity, and oxygen percentage on ignition delay times and ignition limit. These normal-gravity ignition tests have been conducted using the variable oxygen tunnel at Case Western Reserve University. Two ignition times were identified: the first appearance of igniter-assisted gas flame and the minimum igniter energizing time for self-sustained solid ignition. Two types of PMMA samples were used--a circular cylinder and a GEL sphere. The commercially available cylinders were tested first before the custom-made sphere samples became available. Un-instrumented spheres were used for the majority of tests. TC-instrumented samples were employed later only when solid temperature data is needed. The results of the ignition tests have been reported previously in SoFIE monthly meetings. Ignition of two-step heating using hot wire has also been tested. In this series, a low-current heating was first applied until one embedded thermal couple reaches a pre-determined temperature before the higher ignition current is employed. The objective is to test whether an additional preheating method will provide useful insight on the ignition, extinction, or flame spread. Using heat flux sensors, heat flux to the sample surface was measured using cylindrical sample. The variables in these tests are igniter distance, flow velocity, and pressure. The igniter current is kept at 4A.
Bibliography Type:	Description: (Last Updated: 09/19/2025)
Articles in Peer-reviewed Journals	Stalcup EJ, T'ien JS, Jordan J, Wu Z, Nastac G, Li C. "Upward flame spread and extinction over wavy solids." Combustion Science and Technology. 2020;192(7):1173-98. https://doi.org/10.1080/00102202.2020.1738411 , Sep-2020